

Epistasis limits stepwise strain engineering: a complete three-factor test in fatty acid production (13/15 words)

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Abstract (≤ 150 words)

Strain engineering proceeds one edit at a time, keeping what improves and discarding what does not. The procedure assumes that an improving combination can be reached through improving intermediates. We test that assumption on a complete three-factor design: ten transcription factors deleted in every single, double and triple combination in a free-fatty-acid-producing *Saccharomyces cerevisiae* strain, with all 720 deletion orders measured rather than inferred. Interaction is the rule. On total titer under a multiplicative null, 86 of 119 testable triples deviate at a 5% false discovery rate, and the median deviation moves from **+0.40** at two deletions to **−0.72** at three; each of the five measured species interacts too, and they disagree on the sign. Fifty-one triples exceed the base strain, yet only two sit at the end of a route whose every intermediate also improves, and even the shallowest of a triple's six routes falls a median 8% below the starting titer before recovering. None of the seven interaction networks tested is enriched among the interacting triples.

Keywords: metabolic engineering, epistasis, gene interactions, strain design, combinatorial deletion, *Saccharomyces cerevisiae*

Status: draft of a separate publication, not a working note. This is intended for submission as a standalone perspective once the free fatty acid study it re-analyzes is published, so it deliberately tracks the house style of the TorchCell manuscript ([paper/nature-biotech/editing.pdf](#)): same sn-jnl class, same sn-nature bibliography style, same print-approximate editing geometry, same section status chips. Keep it that way, and when a convention changes in [paper/nature-biotech](#), change it here too.

Target: *Metabolic Engineering*, as a research article. The finding is a statement about how build-and-screen campaigns are run, so the readership that acts on it is the one that designs those campaigns, and the journal already publishes analyses of combinatorial strain data. The re-analysis carries original work, the four nulls, the route enumeration and the model comparison, so it is not a commentary. Fallbacks, in order: *ACS Synthetic Biology*, then *Metabolic Engineering Communications*.

Two things are deliberately unfinished. Authorship is a placeholder and must include the authors of the source study. The bibliography is provisional: the document has no Zotero collection of its own, so `references.bib` is built from the served whole-library export, subset to the keys cited here, and only works already in the library can be cited.

✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 Introduction x

A strain is improved by editing it, measuring it, keeping the edits that helped, and editing again. Every round of that loop is a decision made on the strain in hand, and a round that produced nothing better than its parent is a round whose output is discarded. The procedure is the design-build-test-learn cycle in its most common form, and its cost scales with the number of rounds rather than with the number of edits eventually carried.

The procedure is efficient exactly when the effects of edits compose. If a deletion that raises titer on its own keeps raising titer in the presence of other deletions, then a greedy walk reaches a good combination, and the number of strains that must be built grows linearly with the number of edits rather than exponentially. The assumption is rarely stated because it is rarely testable: a campaign that builds a few hundred strains out of a combinatorial space of millions cannot see the combinations it did not build, and cannot tell a combination that is genuinely bad from one it never reached.

Epistasis is the name for the failure of that assumption. In its quantitative form it is the deviation of a combination’s phenotype from what the phenotypes of its parts predict, and it is measured against a stated null [1]. Systematic genetic interaction mapping in *Saccharomyces cerevisiae* established both the scale of the phenomenon on growth rate and the estimator now used for it: a digenic interaction score defined as the departure of a double mutant’s fitness from the product of the two single-mutant fitnesses [2, 3], and a trigenic score that removes the digenic contributions before asking what the third deletion adds [4]. Those maps were built on fitness in rich medium, and the network they describe is itself environment dependent [5]. Whether their conclusions transfer to a production phenotype is not obvious, because fitness and titer are under different selection, are measured on different instruments, and need not share a null.

Combinatorially complete designs are what turn that question into a measurement, and they exist for other phenotypes: four residues of an enzyme active site [6] and a set of chromatin regulators [7]. Large panels of segregants and of perturbed crosses reach further into the genome without being complete in any combination [8, 9]. Metabolic engineering has the multiplexed editing tools to build such a design in a producing strain [10], and the case analyzed here is one.

What has been missing for a production phenotype is a design in which every combination and every construction order is present. Without one, a claim that a combination was unreachable is indistinguishable from a claim that nobody happened to build it. The free-fatty-acid dataset re-analyzed here supplies exactly that: ten transcription factors deleted in every single, double and triple combination in one producing background, which makes all 720 orders of deleting three genes available as measurements rather than as inferences. The design permits a question that a partial screen cannot ask: not how many combinations interact, which any screen can estimate, but how many good combinations a stepwise campaign could have found.

2 Results x

A complete three-factor design over ten transcription factors

The dataset is a combinatorial deletion series in a free-fatty-acid-producing *S. cerevisiae* strain. Ten transcription factors (FKH1, GCN5, MED4, OPI1, RFX1, RGR1, RPD3, SPT3, TFC7 and YAP6) were deleted in every combination of one, two and three genes, giving 10 single, 45 double and 120 triple deletion strains alongside the undeleted base strain. The base strain is itself an engineered chassis, an FFA-overproducing *S. cerevisiae* carrying *pox1Δ faa1Δ faa4Δ*, and the ten transcription factor deletions are additional to those three, so a triple deletion strain carries six deletions in total. Six quantities were measured per strain: the titers of myristate (C14:0), palmitate (C16:0), stearate (C18:0), palmitoleate (C16:1) and oleate (C18:1), and their sum, referred to as total titer. Every titer below is divided by the base strain’s, so a value of 1 is no change and 2 is a doubling.

Completeness is what makes the design usable for the question at hand. Because every sub-combination of every triple was built, each triple can be reached by six distinct deletion orders, and the intermediate strain at each step of each order was measured rather than modeled. The design therefore contains 720 fully measured construction routes.

Only one of the ten single deletions raises titer at all. Deleting GCN5 gives 1.06×; the other nine lose ground, and deleting RPD3 costs nearly half the titer at 0.52×. Combinations behave very differently: 29 of the 45 doubles and 51 of the 120 triples exceed the base strain, and the best strain in the design is a triple at 2.05× (Fig. 2a). The gap between what single deletions do and what combinations do is the first sign that the effects are not composing.

Interaction is the rule, and the third deletion is where it bites

We scored each combination against a multiplicative null, in which the expected phenotype of a combination is the product of its parts’ phenotypes. For a double deletion the digenic score is the deviation from the product of the two

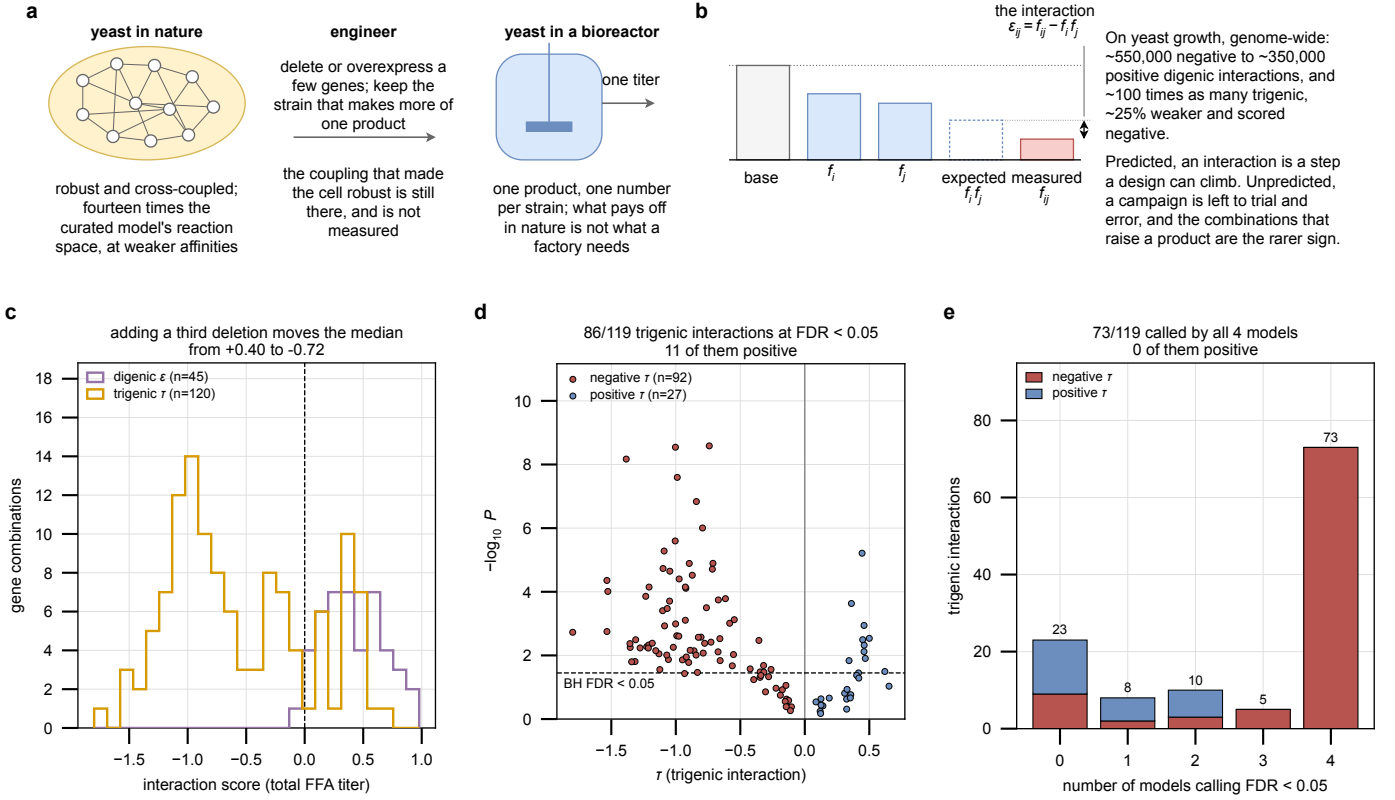


Fig. 1 Interaction is the rule on total fatty acid titer, and the third deletion is where it bites. **a**, Why interactions matter for a factory. A yeast in nature is robust and cross-coupled, and most of its chemistry is off the curated map: retrobiosynthesis with predicted enzyme annotation places 55,734 reactions beside the 4,131 of Yeast9, with median predicted Michaelis constants about twofold higher in that underground network than in the known one [11]. A campaign deletes or overexpresses a few genes and keeps the strain that makes more of one product; the coupling that made the cell robust is still there and is not measured. **b**, An interaction is the departure of a measured combination from an expectation, drawn for the multiplicative null with a negative departure, the common case on growth: genome-wide, about 550,000 negative and 350,000 positive digenic interactions [2], and an estimated hundred times as many negative trigenic ones, about 25% weaker [4]. Predicted, an interaction is a step a design can climb; unpredicted, a campaign is left to trial and error. **c**, Interaction scores against the multiplicative null for all 45 double (lilac) and 120 triple (amber) deletions, as outline histograms on shared bins so the heavy overlap stays readable. The median moves from +0.40 at two deletions to -0.72 at three. **d**, Trigenic interaction score τ against evidence for the 119 testable triples under the multiplicative model. The dashed line marks the largest P that Benjamini-Hochberg rejects at a 5% false discovery rate over the 119 tests of this readout ($P = 0.036$); 86 triples clear it and 11 of those have $\tau > 0$. **e**, How many of the four epistasis models call each triple significant, split by the sign of τ . Seventy-three triples are called by all four with the sign agreeing, and none of the 73 is positive.

singles,

$$\varepsilon_{ij} = f_{ij} - f_i f_j, \quad (1)$$

and for a triple the trigenic score removes the digenic contributions before asking what remains:

$$\tau_{ijk} = f_{ijk} - f_{ij}f_k - f_{ik}f_j - f_{jk}f_i + 2f_i f_j f_k. \quad (2)$$

This is the trigenic score of the yeast genetic interaction maps [3, 4], applied to titer in place of fitness. Substituting Eq. 1 into Eq. 2 rewrites it as a residual against the pairs,

$$\tau_{ijk} = f_{ijk} - f_i f_j f_k - (\varepsilon_{ij} f_k + \varepsilon_{ik} f_j + \varepsilon_{jk} f_i), \quad (3)$$

an identity rather than an approximation. Read this way, τ is what the third deletion adds once the singles and the pairs have been given their due, and each pairwise deviation enters weighted by the remaining single-deletion phenotype. Standard errors propagate through Eq. 2 by the delta method, and each t statistic is referred to the Welch-Satterthwaite effective degrees of freedom of that linear combination, which has a median of 4.28 across the 714 testable trigenic scores.

Deviations from the null are large and are mostly negative at three deletions. The median digenic score is +0.40 and the median trigenic score is -0.72 (Fig. 1c), so adding a third deletion does not damp the interaction, it reverses its typical direction. Of the 119 triples with a testable standard error, 86 depart from the null at a 5% false discovery rate computed over the 119 tests of this readout, and 11 of those 86 depart upward (Fig. 1d).

The pairs are worth stating on their own, because they are what Eq. 3 measures the triples against. On total titer 43 of the 45 doubles have a positive digenic score, 25 clear the same false discovery rate and all 25 of those are positive, and

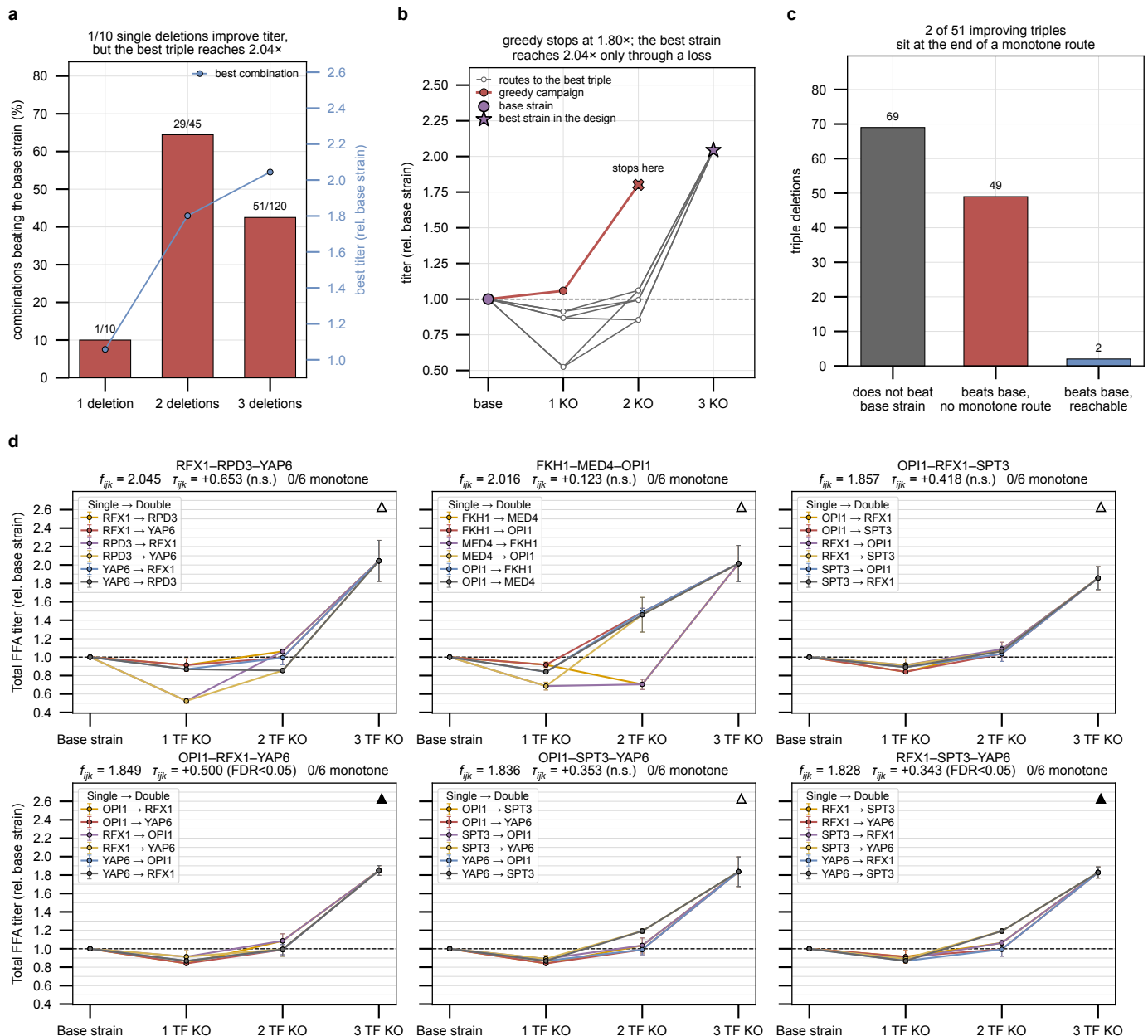


Fig. 2 Improving combinations exist, and almost none of them is reachable one deletion at a time. **a**, Percentage of combinations exceeding the base strain at one, two and three deletions (bars, counts above), with the best titer at each order (blue line, right axis). One of ten single deletions improves titer; the best triple reaches 2.05×. **b**, A greedy campaign (brick) takes the best strictly improving single deletion each round and stops when none improves. It reaches 1.80× after two deletions and halts. Gray lines are all six deletion orders of the highest-titer triple (RFX1, RPD3, YAP6; star), each of which passes through an intermediate below the base strain. **c**, The 120 triples classified by whether they beat the base strain and whether any of their six orders improves at every step. **d**, Trajectories for the six highest-titer triples, one colored line per deletion order, on a shared vertical scale. Rung 0 is the *pox1Δ faa1Δ faa4Δ* chassis and the three rungs after it add transcription factor deletions. Each panel gives that triple's final titer f_{ijk} and its trigenic score τ_{ijk} , and the badge marks the sign and significance of τ . None of the six has a monotone route. Error bars are standard errors of the strain means.

93 of the 120 triples produce less titer than their own singles and pairs predict (Supplementary Note 1). The reversal at the third deletion is therefore a change of direction rather than the continuation of a trend.

Improving combinations are not reachable one deletion at a time

An interaction score says a combination is surprising. It does not say whether a campaign could have found the combination, and that is the question a strain engineer is actually asking. We therefore scored each of the 720 measured routes for monotonicity: a route is monotone when every deletion in it leaves the strain strictly more productive than the strain before it, which is the condition under which a build-and-screen campaign that discards non-improving rounds would follow the route to its end.

Two of the 720 routes are monotone. Relaxing the criterion to allow a step whose loss is within one standard error raises the count to 11 routes across 9 triples, so the result is not an artifact of measurement noise sitting just the wrong

side of zero. Counting by destination rather than by route: 51 of the 120 triples beat the base strain, and 2 of those 51 sit at the end of a monotone route (Fig. 2c). The other 49 improvements exist in the design and are invisible to a stepwise campaign.

The obstacle is a valley rather than a plateau. Every route has a worst intermediate, and how far that intermediate falls below the base strain is the loss a campaign has to accept to follow the route to its end. Each triple has six routes, so it has a best case and a worst case. Taking each triple’s shallowest route, the one whose worst intermediate is highest, the median dips 8% below the starting titer and the deepest such best case dips 22%: no choice of deletion order avoids a loss. Taking the deepest route of each triple instead, which is what a campaign that picks badly gets, the median dips 31% and the worst 48%. A campaign built to discard anything worse than its parent discards the route at the first step either way.

Simulating the campaign directly gives the same answer in one number. Starting from the base strain and taking the best strictly improving single deletion each round, with full visibility of every candidate at every round, the campaign deletes GCN5 ($1.06\times$), then TFC7 ($1.80\times$), and then stops, because all eight remaining third deletions lower the titer (Fig. 2b). The best strain in the design is $2.05\times$, and reaching it requires passing through an intermediate at $0.91\times$ on its most forgiving order and $0.52\times$ on its least. The simulated campaign is more capable than a real one in every respect that matters, and it still recovers 88% of the achievable titer while stopping two rounds short of the answer.

The greedy trap has a second form worth naming, because it is not the same failure. One of the two monotone routes ends at FKH1, GCN5, MED4 at $1.73\times$. A campaign following it would improve at every round and finish with a triple. The simulated campaign does not find it, because at the second round TFC7 ($1.80\times$) beat FKH1 ($1.53\times$) and the better local step led to a dead end. Being reachable is necessary for a stepwise campaign to find a combination, and it is not sufficient.

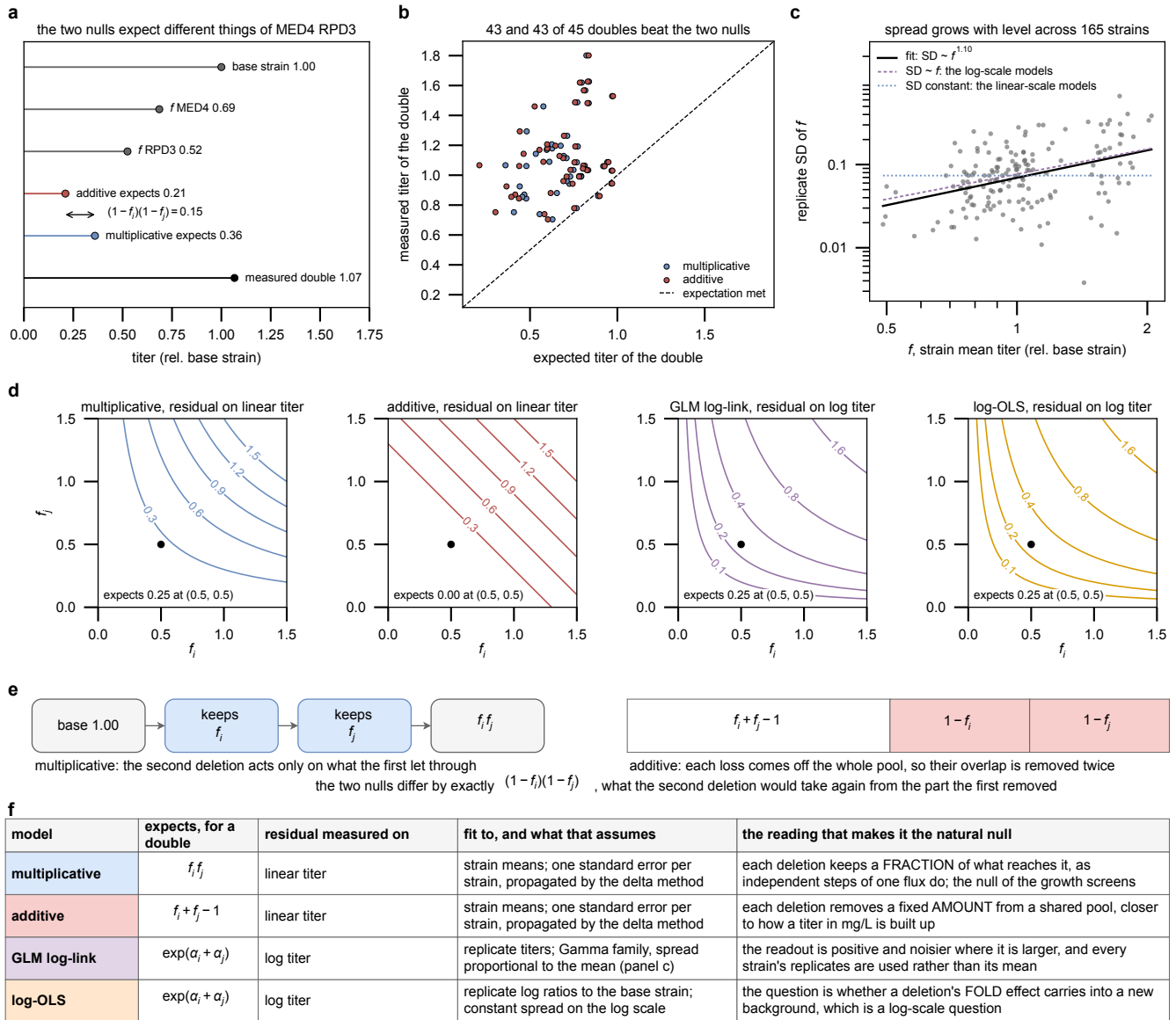


Fig. 3 What each null expects, and what choosing one commits to. **a**, One measured double deletion on a titer axis, MED4 and RPD3, the pair whose two expectations are furthest apart. The additive and the multiplicative expectation differ by $(1 - f_i)(1 - f_j)$, which is an identity rather than a property of this pair. **b**, All 45 doubles against both expectations; distance from the dashed line is the digenic interaction under that model. Forty-three of the 45 sit above both, by a median +0.40 against the multiplicative expectation and +0.45 against the additive one, so the two nulls agree on the direction at two deletions and differ in magnitude. **c**, Replicate standard deviation of f against f over the 165 strains with a nonzero replicate spread, on log axes. The fitted line has slope 1.10 in the logarithms; the two reference lines are the noise assumptions the models divide over, a spread proportional to f , which is the Gamma family behind a log link, and a constant spread. **d**, What each model expects of a double deletion, as level sets over the square of single-deletion phenotypes, one panel per model in the order of **f**. The square runs to 1.5 because single deletions in this design raise titer as well as lower it. Each panel's contours are evenly spaced on the scale that model measures its residual on: at 0.3 to 1.5 in steps of 0.3 for the two linear-scale models, and a doubling apart from 0.1 to 1.6 for the two log-scale ones, which is why those crowd near 1 and open out at low titer. Three of the four expect the same surface, and the additive null is the one that does not: its level sets are straight where the others are hyperbolas. The two surfaces differ by $(1 - f_i)(1 - f_j)$, which is 0.25 where both singles halve the titer; that point is marked in every panel with what the model expects there. **e**, The two families of null as statements about the system. A multiplicative null says each deletion keeps a fraction of what reaches it, which is how independent steps of one flux compose and is the convention the growth interaction screens use. An additive null says each deletion removes an amount from a shared pool, and so counts the part of the first deletion's loss that the second would have taken again. **f**, The four models fit here: what each expects, the scale its residual is measured on, what it is fit to, and the reading of the system under which it is the natural null. Equations in the Methods.

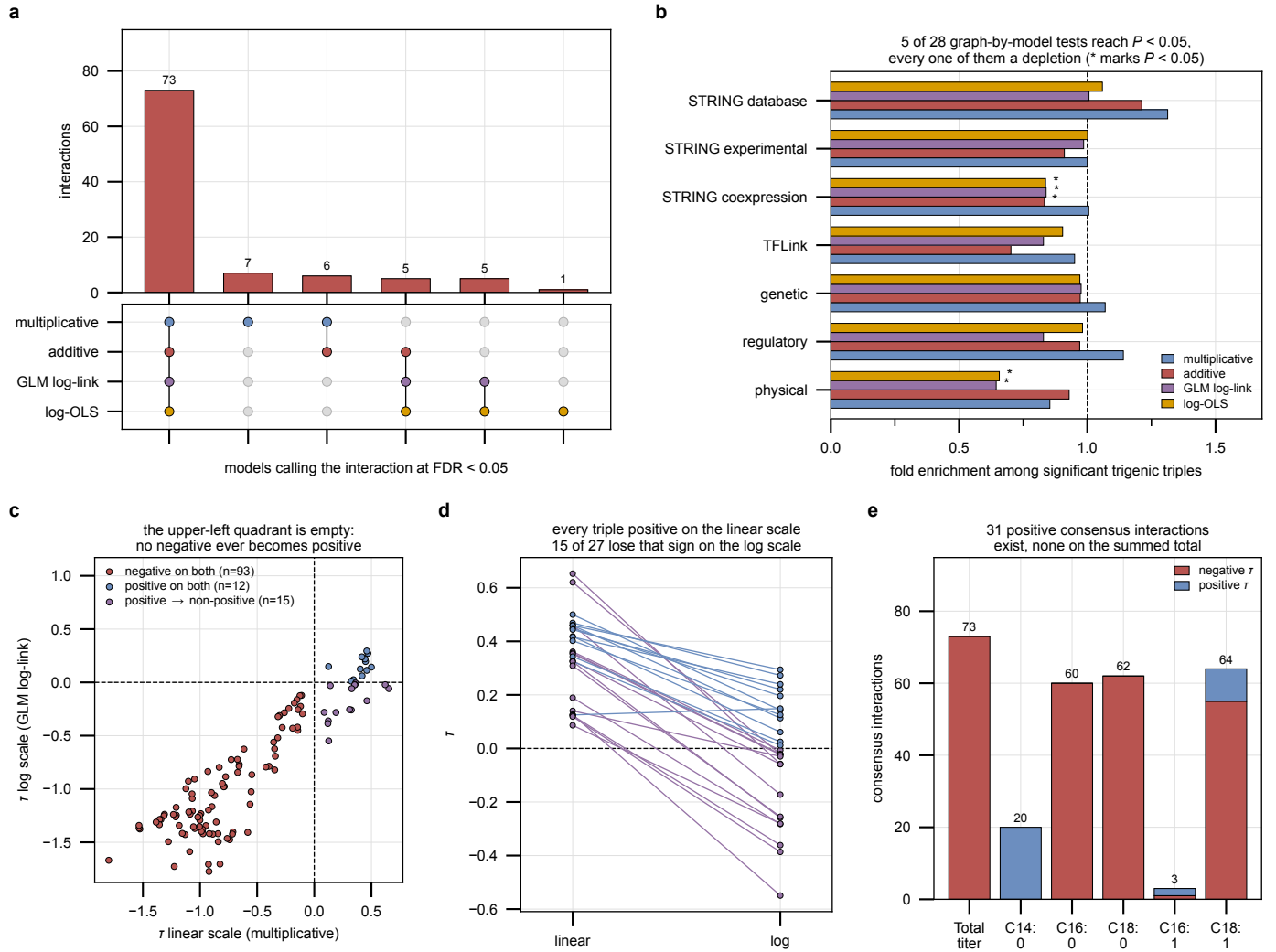


Fig. 4 What counts as an interaction depends on the null, on the scale and on the readout. a, Triples called significant at a 5% false discovery rate by each combination of the four models. Ninety-seven triples are called by at least one model and 73 by all four. **b**, Fold enrichment of seven interaction graphs among the significant trigenic triples, per model, over all six readouts pooled at $P < 0.05$. Asterisks mark $P < 0.05$ by Fisher's exact test; all five significant results are depletions. **c**, Trigenic score on the linear scale against the same quantity on the log scale. The upper-left quadrant is empty, so no combination that is negative on one scale is positive on the other. **d**, The 27 triples with a positive linear-scale score, carried onto the log scale; 15 lose the sign. **e**, Consensus interactions per readout, split by sign. Thirty-one positive consensus interactions exist across the five species, and none of them survives on their sum.

What counts as an interaction depends on the null and on the scale

Equation 2 is one choice among several, and a claim that rests on it should say how much of the claim is the choice. We fit three alternatives on the same normalized strain means: an additive null, in which expected phenotypes add rather than multiply; a generalized linear model with a log link and a Gamma family; and ordinary least squares on log titer differenced against the base strain. The four are not four estimates of one quantity. They encode different statements about how two deletions are expected to combine, and they measure their residuals on different scales (Fig. 3). A multiplicative null says a deletion keeps a fraction of what reaches it, which is how independent steps of a single flux compose and is why the growth interaction screens use it; an additive null says a deletion removes an amount from a shared pool, which is the reading a titer in milligrams per liter invites. The two differ by exactly $(1 - f_i)(1 - f_j)$ at two deletions, so they separate most where both single deletions are severe (Fig. 3a,d).

The four models agree on which combinations interact far more than they agree on the sign. Ninety-seven triples are called by at least one model at a 5% false discovery rate within the readout, 73 by all four, and every one of the 73 is negative (Fig. 4a). The disagreement falls almost entirely on the positive calls, and it falls along the scale: the multiplicative and additive models report 11 and 6 positive triples on total titer, and the two log-scale models report none.

That split is not a difference of power, it is a difference of estimand. The log-scale three-way interaction $g_{ijk} - g_{ij} - g_{ik} - g_{jk} + g_i + g_j + g_k$, with $g = \log f$, agrees with the linear-scale score in magnitude ($r = 0.91$, Spearman $\rho = 0.84$) but not near zero. Of the 27 triples with a positive linear-scale score, 15 are non-positive on the log scale, while no negative linear-scale score becomes positive (Fig. 4c,d). Positive three-way epistasis on fatty acid titer is a linear-scale statement, and reporting one without naming the scale is reporting the modeling choice as a result.

The readout matters as much as the scale. Thirty-one positive consensus interactions, over 29 distinct triples, clear all four models with the sign agreeing on one of the five individual species, and 20 of the 31 are on C14:0. None survives on the summed total (Fig. 4e). Summing the five species is a modeling choice too, and it is the choice that removes the positive signal. The five species behind the sum are not variations on one pattern: the number of triples called within a species runs from 3 to 80, C14:0 reverses sign between the digenic and the trigenic layer, and the summed total sits at $r = 0.91$ from C16:0 and C18:1 but at $r = -0.28$ from C14:0 (Supplementary Note 2). The four expectations themselves, and what each one costs to adopt, are written out in the Methods.

No interaction network on hand predicts which combinations interact

If interactions were predictable from prior knowledge, a campaign could avoid the problem by choosing combinations it expects to be additive. We tested that directly by asking whether the interacting triples are enriched for connectivity in existing interaction graphs. Seven graphs cover these ten genes with at least one connected triple: physical interactions, regulatory interactions, genetic interactions, TFLink, and the coexpression, experimental and database channels of STRING 12.0. For each graph and each model we compared the fraction of significant triples whose three genes form a connected subgraph to the same fraction among the non-significant triples.

Nothing is enriched. Across 28 graph-by-model tests, 5 reach an uncorrected $P < 0.05$ and every one of the 5 is a depletion rather than an excess (Fig. 4b); none of the 5 survives a Benjamini-Hochberg correction over the 28 tests. The largest excess anywhere is the STRING database channel under the multiplicative model at 1.31-fold, which does not reach significance ($P = 0.13$). Drawing the significant interactions onto the pathway itself makes the same point visually: the interacting triples span the transcription factor set with no visible relation to which factors regulate which fatty acid genes (Fig. 5).

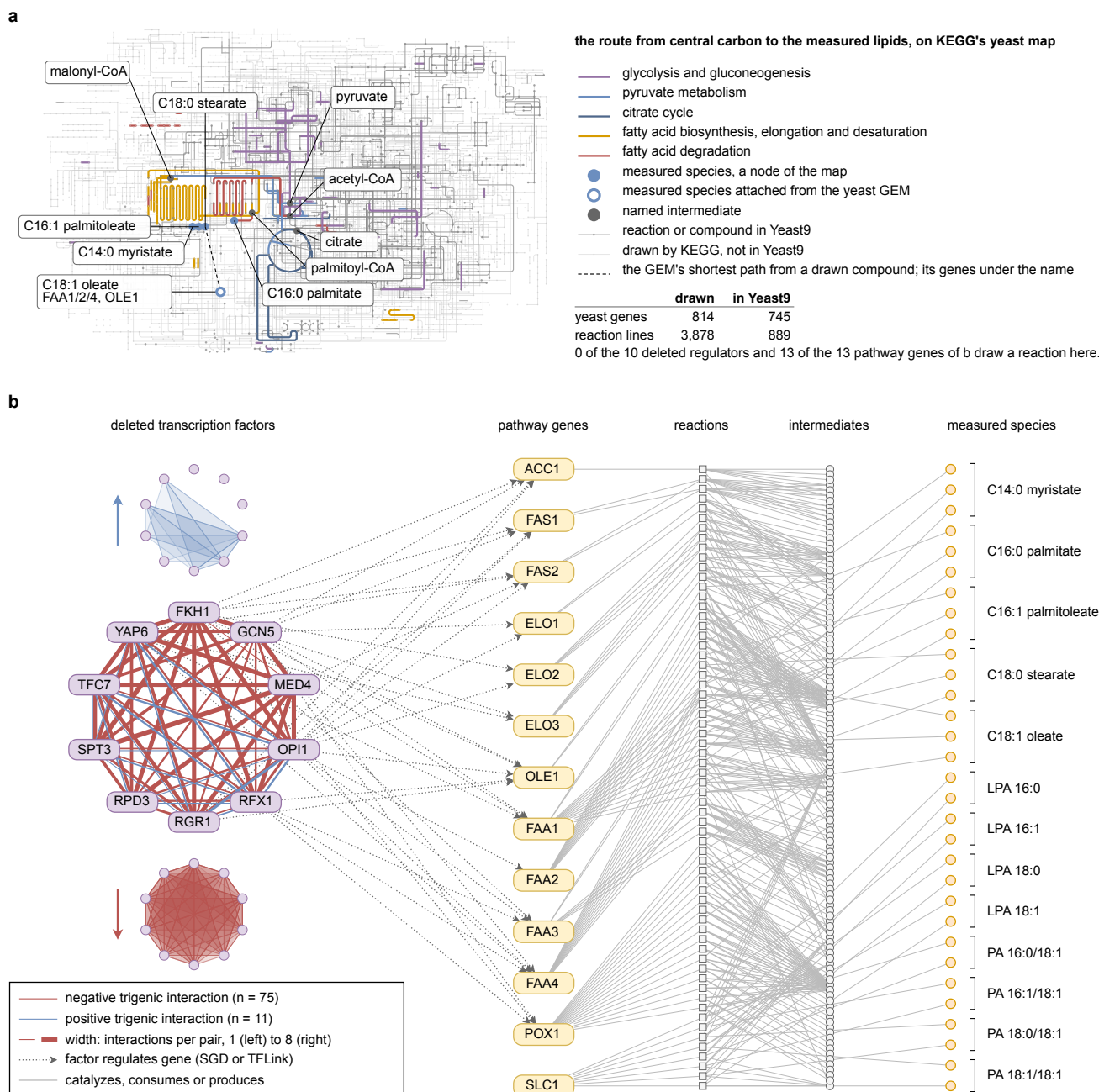


Fig. 5 Trigenic interactions on total fatty acid titer, drawn upstream of the pathway they act through. **a**, Where the measured chemistry sits in metabolism: KEGG's global map of yeast metabolism, redrawn as vector from the coordinates KEGG publishes for it, with the yeast genome-scale model overlaid. Reactions and compounds the model contains are drawn in gray; what KEGG draws but the model lacks is fainter. The map places 814 yeast genes, 745 of them in the model, and 889 of its 3,878 reaction lines; the model's transport, exchange and per-chain lipid reactions have no place on any such map and are the part panel **b** carries. Colored are the yeast reactions of glycolysis and gluconeogenesis, pyruvate metabolism, the citrate cycle, fatty acid biosynthesis, elongation and desaturation, and fatty acid degradation, taken from KEGG's own gene lists for those pathways rather than chosen here. A gene on more than one list takes the first in the order central carbon, degradation, biosynthesis, which is what keeps the degradation arm visible: KEGG lists POX1, the acyl-CoA oxidase this chassis deleted, with the unsaturated fatty acids too. Filled blue nodes are the four measured species the map draws; gray nodes are five named intermediates on the route. Oleate has no node on the map, as the free acid or as oleoyl-CoA, and is attached as an open ring along the shortest path of the same genome-scale model from stearate, with the genes of that path under its name. Of the ten deleted regulators, none is among the genes the map draws, which is the point of the panel: the deletions act on metabolism from outside it. All thirteen pathway genes of panel **b** are drawn. The map's own labels are not reproduced, because at this size they would print below the minimum type size. **b**, The ten deleted transcription factors are arranged in a circle at left. Edges within the circle are the 86 trigenic interactions significant at a 5% false discovery rate within the total-titer readout under the multiplicative model, 75 negative (brick) and 11 positive (blue), each drawn as the three edges of its triangle; edge width is the number of interactions the pair takes part in, from 0.40 pt at 1 to 2.15 pt at 8. Every one of the 45 pairs is in at least one negative interaction, so a pair that also carries a positive one would hide it: the two signs of a pair are therefore drawn as two parallel lines, offset perpendicular to the pair by half their combined width, and neither covers the other. Above and below that circle the same ten factors are drawn again at the same angles, unlabeled, carrying only the positive interactions (blue, above) and only the negative ones (brick, below). The two shapes are what the combined circle cannot show. The negatives reach all 45 pairs and every factor, with up to 8 interactions on one pair. The positives reach 17 pairs and 7 factors, at most 4 on a pair, and FKH1, GCN5 and MED4 take part in none of them. Dotted arrows are regulatory links from a factor to a pathway gene in the SGD regulatory graph or TFLink; three of the ten factors have none. The columns to the right are the 13 core fatty acid pathway genes, the 64 reactions they catalyze, the intermediates those reactions consume or produce, and the 31 measured species, one node per compartment (peroxisome, ER membrane, lipid particle) with a bracket per species. LPA, 1-acyl-sn-glycerol 3-phosphate; PA, phosphatide, written as the two acyl chains.

A model trained on growth interactions does not predict the titer interactions

The other place a campaign could look for a predictor is a model trained on measured combinations rather than on a graph. The torchcell equivariant cell graph transformer, described in a companion manuscript in preparation ([paper/nature-biotech/editing.pdf](#)), is trained on the trigenic growth interaction screens of Kuzmin et al. [4, 12], where each triple deletion is scored by colony fitness in a wild-type background. What that model is measured against, and how it behaves on the additive nulls, is worked through in [13]. All ten factors here are in its gene space, and seven of them were deleted somewhere in its training data; GCN5, RPD3 and TFC7 were not. We scored each of the 120 triples as a three-gene deletion and asked whether the predicted growth interaction tracks the measured titer interaction under the multiplicative model (Fig. 6). The two quantities differ in readout and in background, so a null here does not separate a model that cannot rank trigenic interactions from a growth interaction that does not carry over to titer. The question is only whether this predictor, as it stands, would help choose combinations.

Neither magnitude nor rank transfers. Across the 120 triples the ensemble mean of three checkpoints correlates with the measured τ at $r = 0.16$ ($P = 0.08$) and Spearman $\rho = 0.13$ ($P = 0.17$), and the measured τ shows no trend across quartiles of the prediction (Kruskal-Wallis $P = 0.39$). The three checkpoints agree with each other (r 0.62 to 0.71) far better than any of them agrees with titer (r 0.08 to 0.20), so the miss is a property of what the model learned rather than of one training run. Restricting to the 86 triples called at a 5% false discovery rate, to the 84 free of the contested TFC7 locus, or to the 35 whose three factors were all deleted in training leaves r between 0.12 and 0.15, no further from zero than the full set, and the top quartile by predicted τ holds no more called interactions than the rest (22 of 30 against 64 of 90).

The one signal is the sign. Of the 20 triples the model predicts positive, 9 measure positive, against 18 of the 100 it predicts non-positive: 45% against 18% (Fisher $P = 0.016$). The same split holds in the 84 triples without TFC7 ($P = 0.005$) and in the 35 fully trained ones ($P = 0.04$). Every P in this subsection is nominal among ten related tests on one set of 120 triples, so the sign association is a lead to test on a new panel, not a result. Read with the graph result, the training data a useful predictor needs are combinations measured on the readout of interest: a model trained on growth combinations carries over to titer at most the direction of the interaction.

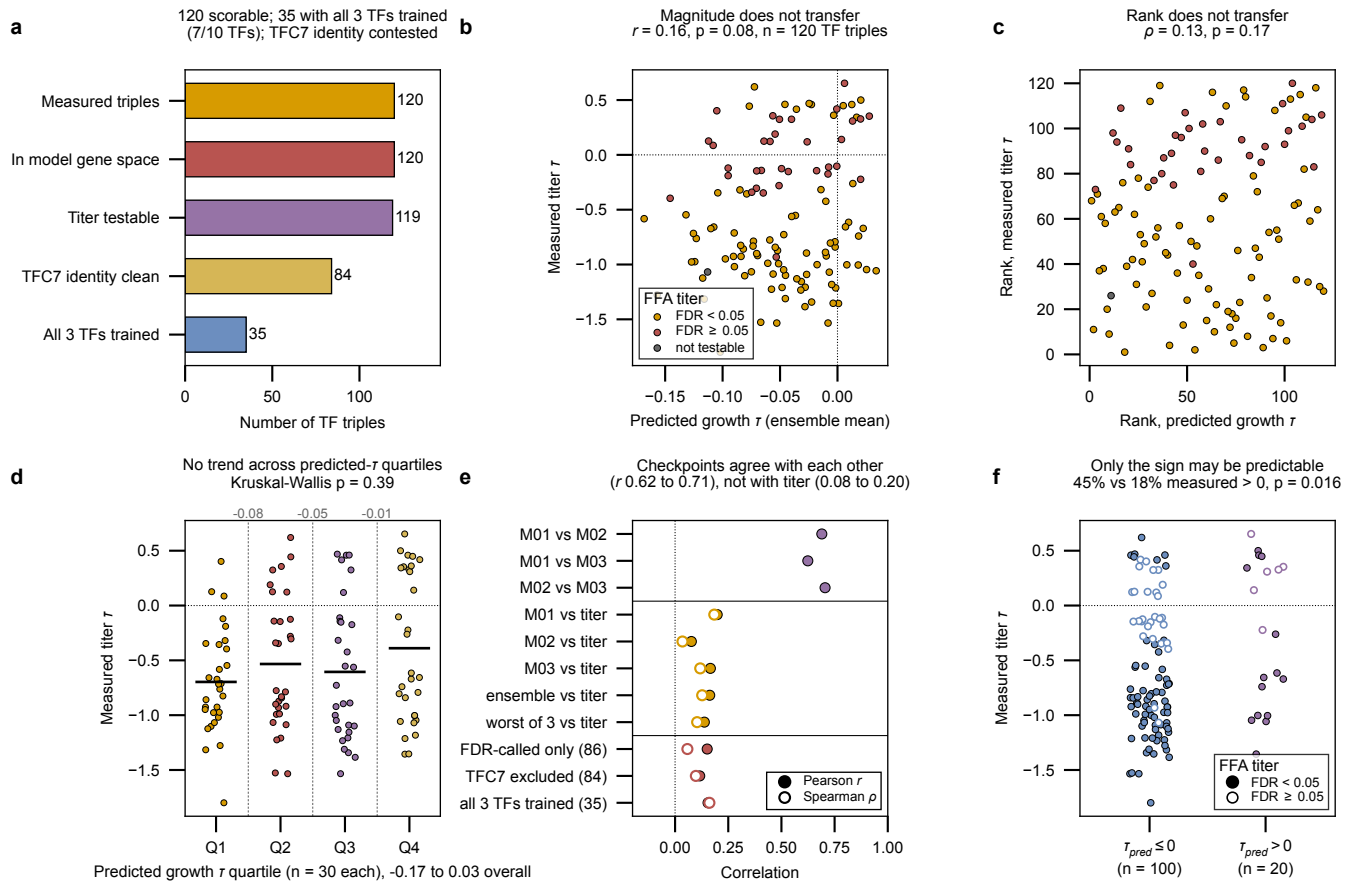


Fig. 6 A growth-trained transformer does not predict the titer interactions. Three checkpoints of the torchcell equivariant cell graph transformer, trained on the Kuzmin 2018 and 2020 trigenic growth screens, scored each transcription factor triple as a three-gene deletion in a wild-type background; the measured quantity is the multiplicative titer interaction τ of the same triple on the *pox1Δ faa1Δ faa4Δ* chassis, total titer. “FFA titer FDR” is Benjamini-Hochberg over the 119 testable triples within total titer. **a**, Coverage: 120 scorable, 35 with all three factors seen as deletions in training (7 of 10 factors), 84 free of the contested TFC7 locus. **b**, Ensemble-mean predicted growth τ against measured titer τ , Pearson $r = 0.16$, $P = 0.08$; the key’s classes number 86, 33 and 1. **c**, The same as ranks, Spearman $\rho = 0.13$, $P = 0.17$. **d**, Measured τ by quartile of predicted τ , bar at the bin mean, Kruskal-Wallis $P = 0.39$. The dashed dividers are the three cut points between quartiles, labeled with the predicted τ at each. **e**, Pearson (filled) and Spearman (open) correlations: between checkpoints on the same triples; each checkpoint, the ensemble mean and the worst of three against titer; the ensemble on three subsets. **f**, Measured τ split by predicted sign (blue at or below zero, purple above; filled where the titer interaction is FDR-called), 45% against 18% measured positive, Fisher $P = 0.016$. Every P is nominal among ten related tests on one set of 120 triples.

3 Discussion x

The design space of a strain engineering campaign is usually described as too large to search. Ten deletable targets already give 1,023 combinations, and a genome-scale target list gives more combinations than there are strains anyone will ever build. That framing puts the difficulty in the size of the space and implies that the remedy is throughput or a better search.

The measurements here point at a different difficulty, which throughput does not fix. In a space small enough to enumerate completely, with every combination and every construction order measured, the best strain is still unreachable by the procedure that campaigns actually use. The obstacle is not that the good combination is hard to find among many. It is that the good combination sits behind an intermediate that a screening rule would throw away. Of 51 improvements, 2 are reachable. Adding throughput multiplies the number of routes a campaign can try; it does not make a route monotone.

Three consequences follow for how a campaign is designed.

Retain losses. A rule that keeps only strains better than their parent discards the approach to almost every improvement in this design. Carrying a fixed number of non-improving intermediates forward per round costs build capacity in proportion, and it is the only change to the procedure itself that addresses the reachability problem. How far down to keep looking is set by the depth of the valleys: a median of 8% of the base titer on each triple’s shallowest route, and 31% on its deepest.

Build combinations, not sequences. If the intermediates are uninformative about the endpoint, then building them is a cost with no return, and a campaign is better served by constructing the combinations it wants directly. That

is a statement about what multiplexed editing buys: not speed at reaching a known target, but access to targets that no sequence of single edits reaches.

Predict the combination, not the parts. A model that predicts a combination’s phenotype from its parts’ phenotypes fails on exactly the combinations that matter here, because those are the combinations where the parts do not compose. A useful predictor has to be trained on combinations, and the enrichment result says the existing interaction graphs will not stand in for that training data: across 28 graph-by-model tests, the interacting triples are never enriched for connectivity in any graph, and are depleted in five at an uncorrected $P < 0.05$. Whatever determines which combinations interact is not recorded in physical, regulatory, genetic, TFLink or STRING connectivity among these ten genes. Nor is it carried by combinations measured on a different readout: a transformer trained on trigenic growth interactions predicts neither the magnitude nor the rank of the titer interactions, and at most their sign. The combinations a predictor learns from have to be measured on the phenotype it is asked to predict.

What theory predicts about where epistasis comes from

There is a body of theory on when deletions in a metabolic system should compose and when they should not, and it is worth asking what it predicts here. Flux balance analysis over the yeast metabolic network gives a trimodal distribution of pairwise epistasis and organizes the interactions into modules that interact monochromatically, meaning that the links between two functional modules are either all buffering or all aggravating [14]. A complementary result derives how epistasis propagates in a hierarchical metabolic network with first-order kinetics, and finds that weak epistasis at a lower level is distorted as it moves to higher-level phenotypes, so pairwise interactions between genes should be common and should depend on the genetic background [15]. Both predict what is observed here at the level of prevalence. Interaction is common, and its sign is consistent within a readout.

What neither predicts is the reachability result, and the reason is that both are theories about enzymes in a flux network rather than about transcription factors upstream of one. The first says so in its own words, excluding “beneficial mutations, partial loss of function, regulatory genes or multicellular developmental networks” from its scope [14]. A factor deletion changes the expression of many pathway genes at once, and it also changes genes outside the pathway, so the module structure a flux model assigns to the pathway is not the structure the deletions act on. The one systematic study of epistasis between gene-specific transcription factors in yeast reached its mechanisms through expression profiling of the double deletions rather than through pathway topology [16], and proposed two routes, buffering by induced dependency and alleviation by derepression, that connect genes in parallel pathways with no shared reaction. The enrichment result here is consistent with that: connectivity in a physical, regulatory, genetic or coexpression graph does not predict which of these triples interact. Applying the metabolic theory to a factor panel would require a model that maps a factor deletion to a distribution of flux changes first, and that model is what the field does not yet have for titer.

Why a regulator panel might interact more than an enzyme panel would

The interactions reported here are dense, and the perturbations that produced them are regulators rather than enzymes. Whether those two facts are connected is worth separating into what the literature supports and what it does not, because the connection is easy to assert and has consequences for which panels a campaign should expect trouble from.

Two things are established. The first is that an interaction score measured several steps away from where a mutation acts does not require the mutations to interact. Nonspecific epistasis, which arises from the curvature of the map between the layer a mutation acts on and the layer that is assayed, applies to every combination rather than to particular ones, and it is scored as specific and higher-order interaction whenever the curvature is not modeled [1]. In a hierarchical metabolic network with linear kinetics, epistasis at a higher-level phenotype is forced even when there is none at the lower-level one, because the map between the two is nonlinear [15]. The second is that a single scalar readout aggregates contributions that are not separately measured, and that a large fraction of an apparently rugged landscape can be the aggregation rather than the biology: in a combinatorially complete enzyme landscape, a single global nonlinearity accounts for three quarters of the variance in fitness [6]. A titer in milligrams per liter is exactly such an aggregate.

The step from there to “a regulator interacts more than an enzyme” is a hypothesis, and it is ours rather than the literature’s. Stated precisely, it has three parts: a regulator acts on the measured flux through more intervening reactions than an enzyme of that flux does, each of which can contribute curvature; a regulator changes many reactions at once, most of them not measured here, and much of what a cell does is not on the curated map at all, since retrobiosynthesis with predicted enzyme annotation places 55,734 reactions beside the 4,131 of Yeast9, at predicted affinities about twofold weaker than the known network’s [11]; and one measurement at one metabolic locus therefore reports the combined effect of many small unmeasured changes, which is scored as an emergent deviation. None of the three is tested in the literature cited here, and the closest measured statements cut both ways. The mixed-epistasis argument that more intervening steps give “additional possibilities for interconnectivity” is made about kinases against

transcription factors, with a transcription factor as the short-distance case, because the readout there is expression rather than a metabolite [16]. Genetic interaction degree does track multifunctionality and pleiotropy, and chromatin organization and transcription are named hub processes of the nonessential network [2], which supports the prevalence half; it is measured on fitness, and lipid metabolism is itself in the high-pleiotropy bracket of the same analysis, which is awkward for a fatty-acid panel in particular.

What the hypothesis must not assume is that enzymes are the additive baseline. They are not. The flux through a pathway is a concave saturating function of any one enzyme’s activity rather than a linear one [1], so two enzymes of one pathway need not compose even when nothing connects them but the flux; in the hierarchical treatment a mutation in any enzyme interacts with a mutation in any other enzyme of its module [15]; and the combinatorially complete landscape in which epistasis was measured is four residues of one active site, which is no network distance at all [6]. A regulator-against-enzyme claim has to beat that baseline rather than assume it away. The sign is open too. Where single-gene knockdowns were crossed into segregants of differing fitness, the fitter backgrounds were the more sensitive to a perturbation, so the cost of a perturbation depends on how much headroom the background has [8]. Whether a regulator panel sits at the sensitive end of such a relation is not something this design can say.

Settling this needs a comparison the design here cannot make: an enzyme panel of the same shape, combinatorially complete, in the same chassis, on the same readout. Simulating that comparison instead does not substitute for it. Running both arms through a flux model of yeast metabolism, with a deletion scaling the capacity of every reaction it touches and no interaction term anywhere in the perturbation, produces nothing at all at the capacity multipliers a reader would try first, and below those it produces scores that are arithmetic on strains which have stopped growing rather than a graded response. In that model the regulator arm also sits closer to the readout than the enzyme arm, not further, because a regulator’s reaction set is the union over hundreds of targets (Supplementary Note 3). Reach is not the explanation either, at least not on the annotation graphs. Counting, on nine of the ten gene graphs the enrichment above scans and on the model’s own gene-reaction-metabolite graph, how many genes and how many model reactions lie within one hop of a deleted factor, of a pathway enzyme, and of an enzyme one reaction away from the pathway, the ten factors reach about twice as many genes as the pathway enzymes (median 1,416 against 701) but no more reactions outside the measured locus (904 against 975, Mann-Whitney $P = 0.16$), and at two hops every class reaches all but one of the model’s gene-annotated reactions outside that locus (Supplementary Note 4). What separates the classes is not how far they reach but through which graph they reach it: a factor touches metabolism only through its targets, an enzyme through coexpression and association, and the union of the curated graphs is dense enough that the difference is gone one hop later. Whether reach on an annotation graph stands in for effect at the readout is not tested by any of this. Until the measured comparison exists, the honest form of the claim is that interaction was pervasive in a regulator panel measured at one metabolic locus, that a nonlinear map from the perturbed layer to the measured one is sufficient to produce it, and that neither the amount nor the sign is predicted from the panel’s distance to the readout or from its reach.

What this design can and cannot support

The design is complete, which is what licenses the reachability claim, and it is small, which bounds it. Ten transcription factors in one producing background, one medium and one harvest time do not establish a rate at which improvements are unreachable in general. They establish that the rate can be 2 in 51 in a real production campaign’s target set, which is enough to make the assumption of composability something that has to be checked rather than assumed.

Two features of the system may not transfer. The deleted genes are transcription factors rather than pathway enzymes, so a deletion acts on many pathway genes at once and the interactions may be denser than a comparable enzyme panel would give. And the phenotype is a titer under production conditions rather than growth rate, so the deletions are under no selection to be individually tolerable. Both cut toward more interaction here than a reader should expect elsewhere. A single-gene-per-step campaign on enzymes may well find a monotone route more often. What the design shows is that the question has an answer, and that the answer is measurable rather than assumable.

The measurement itself carries one limitation that applies to every number in this document. Each strain contributes an endpoint titer in milligrams per liter from at most three replicate cultures, and the released data include no biomass measurement, so nothing here is normalized to cell density. A deletion that lowers final biomass and a deletion that lowers fatty acid output per cell are indistinguishable in this readout, and both appear as a lower f . Two consequences follow and neither is removable by reanalysis. Interaction scores mix specific productivity with growth, so a triple whose three deletions each cost biomass could show a negative τ with no change in per-cell output. And the valley depths that drive the reachability result are titer valleys, which is the right quantity for a campaign that screens on titer, and is not evidence about the yield or the rate. What the replicates do support is that the values are stable: the trigenic standard errors propagate from up to three replicates per strain through seven strains, and the monotone-route count moves only from 2 to 11 when a step is allowed to lose up to one standard error, so the result is not sitting on measurement

noise. A design that reported optical density alongside titer would separate the two routes, and that is a property of the source experiment rather than of the analysis.

What would settle it

Answering the general question needs more datasets of the same shape, and the shape is restrictive. A partial combinatorial screen cannot support a reachability claim at all, because an unbuilt intermediate is indistinguishable from a bad one. The requirement is a design in which every sub-combination of every reported combination was itself built and measured, in the same background and the same batch. Screens that report only the top performers, or only the combinations that a campaign happened to reach, do not qualify no matter how many strains they contain.

The complementary route is theory. A production phenotype is the output of a metabolic network under constraints, and a network model that predicts where in a pathway deletions compose and where they do not would turn the reachability question into a calculation performed before the strains are built. The pieces exist for enzymes in a flux network [14, 15], and what is missing is the step from a regulator deletion to the flux changes it causes. That model does not exist yet for titer, and the enrichment result above says it will not be recovered from connectivity alone. Building it is the direction this line of work points.

4 Methods ✕

Data and normalization

Titers for five fatty acid species and their sum were taken from the source study’s supplementary tables for the base strain and for all 175 deletion combinations. Each strain contributes up to three replicate measurements per species, in milligrams per liter at a single endpoint: three for all but twelve strains, two for eleven of those twelve, and one for the triple GCN5 OPI1 TFC7, which is why 119 of the 120 triples carry a testable standard error. Every titer is divided by the base strain’s titer for the same species in the same experiment, so all values below are ratios and the base strain is 1 by construction. The sum, referred to as total titer, is formed per replicate before averaging rather than by summing the five published species means, so it is a measured quantity and not a sixth derived one. Loading and normalization are in `experiments/008-xue-ffa/scripts/free_fatty_acid_interactions.py`.

The released measurements carry no biomass or optical density column, so no quantity in this document is normalized to cell number. Every f is a volumetric titer ratio, and a deletion that lowers final biomass is not separable from one that lowers output per cell. This is a property of the released data rather than a choice made here, and it bounds what the interaction scores mean: they are interactions on volumetric titer, which is the quantity a titer-screened campaign selects on.

Epistasis models

An interaction score is a residual, and a residual is only defined once the expectation is written down. The four models below write down four different expectations for the same seven measured strains, so a triple can be an interaction under one and not under another without either fit being wrong. Figure 3 draws the four side by side on one worked pair of deletions.

Setting. Write f_S for the titer of the strain carrying deletion set S , divided by the titer of the base strain in the same experiment, so $f_\emptyset = 1$ by construction. Single, double and triple deletions are f_i , f_{ij} and f_{ijk} . All four models are fit on these same normalized quantities, and none of them sees a combination the design did not build.

Multiplicative. The expected phenotype of a combination is the product of its parts, which is the null of the yeast genetic interaction maps:

$$\varepsilon_{ij}^\times = f_{ij} - f_i f_j, \quad \tau_{ijk}^\times = f_{ijk} - f_{ij} f_k - f_{ik} f_j - f_{jk} f_i + 2f_i f_j f_k. \quad (4)$$

Independence on this scale means that a deletion multiplies titer by the same factor whatever else is deleted. These are Eqs. 1 and 2, and this is the model every figure uses unless it says otherwise (`free_fatty_acid_interactions.py`).

Additive. The same construction with sums in place of products:

$$\varepsilon_{ij}^+ = f_{ij} - f_i - f_j + 1, \quad \tau_{ijk}^+ = f_{ijk} - (f_{ij} + f_{ik} + f_{jk}) + (f_i + f_j + f_k) - 1. \quad (5)$$

Independence here means that a deletion shifts titer by the same amount whatever else is deleted. The +1 and -1 are the base strain entering at $f_0 = 1$ (`additive_free_fatty_acid_interactions.py`).

GLM log-link. A generalized linear model with a Gamma family and a log link, fit on the replicate-level titers rather than on strain means, with $x_i \in \{0, 1\}$ marking deletion i and r indexing replicate:

$$\log \mathbb{E}[y_{S,r}] = \mu + \sum_i \alpha_i x_i + \sum_{i < j} \beta_{ij} x_i x_j + \sum_{i < j < k} \gamma_{ijk} x_i x_j x_k + \rho_r. \quad (6)$$

The interaction is γ_{ijk} . The base strain is excluded from the fit and enters through μ . Gamma sets the variance proportional to the square of the mean, which is the right shape for a titer that cannot go negative and whose spread grows with its level (`glm_log_link_epistatic_interactions.py`).

Log-OLS. Ordinary least squares on log titer differenced against the base strain within each replicate, with $\delta = 10^{-6}$ a pseudocount:

$$g_{S,r} = \log(y_{S,r} + \delta) - \log(\bar{y}_{0,r} + \delta), \quad (7)$$

regressed on the same saturated design as Eq. 6. Its three-way coefficient estimates the saturated log-scale interaction

$$\tau_{ijk}^{\log} = g_{ijk} - g_{ij} - g_{ik} - g_{jk} + g_i + g_j + g_k, \quad g_S = \log f_S, \quad (8)$$

and that equivalence, which Fig. 4c,d rests on, is checked in code rather than assumed: `linear_vs_log_scale_sign_test.py` recomputes Eq. 8 from the strain means and raises unless it agrees with both fitted log-scale models at $r > 0.99$ (`log_ols_wt_differencing_epistatic_interactions.py`).

The scale is the first tradeoff, and it is not a matter of taste. Equations 4 and 8 are different quantities, not two estimates of one quantity. They agree in magnitude across all 120 triples at $r = 0.91$ and Spearman $\rho = 0.84$ and disagree near zero, which is exactly where a call is made: 15 of the 27 triples with a positive linear-scale score are non-positive on the log scale, and none of the reverse. A three-way interaction is not invariant under a monotone transform of the phenotype, so a positive trigenic interaction on titer is a linear-scale statement and has to be reported as one. Which scale is appropriate depends on what the reader will do with the number. A campaign that asks how many milligrams per liter a third deletion adds is asking a linear-scale question; a campaign that asks whether a deletion's fold effect is preserved in a new background is asking a log-scale one.

The second tradeoff is what carries the pairwise information. The multiplicative trigenic score can be written with the digenic scores in it, which is Eq. 3 and is an identity rather than an approximation. Each pairwise deviation enters weighted by the remaining single-deletion phenotype, so a pair that interacts strongly in a background where the third deletion is nearly neutral contributes almost its full ε , and the same pair contributes little when the third deletion is severe. The log-scale interaction in Eq. 8 weights every term equally. Neither weighting is more correct; they encode different ideas of how a pair is expected to act in the presence of a third deletion, and the weighted form is the one that keeps comparability with the published yeast interaction maps.

The third tradeoff is power against assumption. The two linear-scale models are fit on strain means and propagate the replicate standard errors through Eqs. 4 and 5 by the delta method, so they assume nothing about the shape of the noise beyond a standard error per strain, and they pay for it in degrees of freedom: a trigenic score combines seven independently measured strains at three replicates each, giving a median Welch-Satterthwaite effective degrees of freedom of 4.28. The two regression models are fit on the replicate-level values and borrow strength across the whole design, which raises power and buys a mean-variance assumption in exchange, Gamma for Eq. 6 and homoscedastic on the log scale for Eq. 7. A count of significant interactions is therefore not comparable across the two families without saying which is which, and that is why the Results report agreement on the identity of the interacting triples separately from agreement on their sign.

What this document uses, and why. The multiplicative model is the reported one throughout, for comparability with the trigenic screens the field already has, and every claim that rests on a sign is stated with the scale attached. The other three exist here to answer how much of a claim is the modeling choice, which is a question the reader can only settle if all four expectations are on the page.

Uncertainty and multiple testing

Standard errors propagate through the interaction expression by the delta method, using the per-strain standard errors of the replicate means. The resulting t statistic is referred to the Welch-Satterthwaite effective degrees of freedom of the linear combination, computed from each term's contribution to the combined standard error. This matters more than it usually does. A trigenic score combines seven independently measured strains, and referring its t statistic to the degrees

of freedom of the smallest single input, as an earlier version of this analysis did, gives 1 or 2 rather than a median of 4.28, and suppresses nearly every call. Across the 990 multiplicative tests of the whole run, none was significant at a 5% false discovery rate before the correction and 456 are after it (audit record: `notes/plan.008-xue-ffa-epistasis-audit.2026.08.15.md`).

Benjamini-Hochberg controls the false discovery rate. The family matters and is stated wherever a count appears. Counts in Figs. 1 and 4 correct over the tests of a single readout, since that is what those figures claim about. The enrichment analysis in Fig. 4b uses the stored per-test $P < 0.05$ over all six readouts pooled, which is the family the enrichment scripts were written against.

Path enumeration and monotonicity

For each triple, all six orders of introducing its three deletions were expanded into a four-rung trajectory: base strain, one deletion, two deletions, three deletions, taking the measured strain at each rung. A route is monotone when every step is strictly positive, and monotone within one standard error when every step exceeds the negative of that step's propagated standard error. Valley depth on a route is $1 - \min(f_i, f_{ij})$, the shortfall of its worst intermediate, taken over the two intermediate rungs only, since the base strain and the destination are not intermediates. A triple has six routes, so both ends of that set are reported: the minimum over the six is its shallowest approach, the best a campaign with perfect foresight could take, and the maximum is its deepest. The greedy campaign in Fig. 2b starts at the base strain, evaluates every combination reachable by one further deletion, takes the highest if it exceeds the current strain, and stops otherwise. Enumeration and monotonicity are in `ffa_epistatic_path_panels.py`; the greedy simulation is in `perspective_figure_panels.py`.

Interaction graph enrichment

Ten interaction graphs were queried for each triple: physical, regulatory and genetic interactions, TFLink, and the neighborhood, fusion, co-occurrence, coexpression, experimental and database channels of STRING 12.0. A triple counts as overlapping a graph when its three genes induce a connected subgraph of that graph. Fold enrichment is the ratio of the overlapping fraction among significant triples to the overlapping fraction among the rest, and significance is Fisher's exact test on the corresponding two-by-two table. Three of the ten graphs contain no connected triple among these ten genes at all and are omitted from Fig. 4b, which is a statement about those graphs' coverage of this gene set and not about epistasis (`interaction_graph_enrichment_analysis.py`).

Scoring the triples with a growth-trained model

Each of the 120 triples was scored by the torchcell equivariant cell graph transformer trained on the Kuzmin 2018 and 2020 trigenic growth screens, as a three-gene deletion in a wild-type background, the perturbation form the model was trained on. The three chassis deletions were not part of the scored genotype, and no single or double was scored. Three independently trained checkpoints were scored on CPU with a parity guard against stored GPU predictions of the same genotypes (18 values, largest deviation 0.0012 at a tolerance of 0.005), and the prediction is the ensemble mean of the three. The predicted quantity is the trigenic growth interaction τ on the synthetic genetic array selection medium. It was compared with the measured multiplicative titer interaction by Pearson and Spearman correlation, by a Kruskal-Wallis test of measured τ across quartiles of the prediction, by Fisher's exact test of the predicted sign against the measured sign, and by Fisher's exact test of the top predicted quartile against the false-discovery-rate call (`experiments/010-kuzmin-tmi/scripts/ffa_panel_model_score.py`, tables in `experiments/010-kuzmin-tmi/results/ffa_panel_model_score.\{csv,json\}`).

The metabolic map panel

Figure 5a is KEGG's global map of yeast metabolism, redrawn from the coordinates KEGG publishes for it and overlaid with the yeast genome-scale model. No part of it is hand-placed. The generating script is `experiments/008-xue-ffa/scripts/ffa_kegg_map.py` and every response it reads is stored with its sha256 under `experiments/008-xue-ffa/results/kegg_map/`.

Why a reference map and not a layout of the model. A metabolic map that reads as one, with the citrate cycle drawn as a circle and glycolysis as a spine, is a hand-drawn artifact. No layout algorithm produces one from a stoichiometric model, the yeast genome-scale model ships no map, and the only published yeast map in Escher is of an earlier model and covers central carbon alone. What is available is the coordinates of a map somebody already drew.

KEGG publishes its yeast global map `sce01100` as KGML, which carries the vertex list of every reaction line and the center of every compound circle, so the drawing can be taken and the rendering replaced.

What is fetched. Three kinds of response, each cached and hashed, and re-fetched only under `--refresh`, which raises rather than overwrites if the bytes have changed upstream. The KGML of `sce01100` gives 3,878 polylines and 3,562 compound circles on a canvas of 4,858 by 3,137 units. One gene list per pathway (`link/sce/<pathway>`) gives the yeast genes KEGG assigns to each of the seven modules the route passes through. One lookup per named gene (`find/sce/<name>`) resolves a gene name to its systematic name, preferring the entry whose first symbol is that name, which is what distinguishes TFC7 as YOR110W from the YNL039W that carries TFC7 only as an alias.

What is overlaid. The yeast genome-scale model, Yeast9 (yeast-GEM 9.0.2), is read from its SBML with `cobrapy`, and its gene identifiers, its `kegg.reaction` annotations and its `kegg.compound` annotations are cached under the sha256 of the model file. A drawn line counts as being in the model when any gene KEGG attaches to it is a gene of the model or any reaction it carries is one of the model's KEGG reaction identifiers; a drawn circle counts when its compound identifier is one of the model's KEGG compounds. The map draws 814 yeast genes, of which 745 are in the model, and 889 of its 3,878 lines and 684 of its 3,562 circles are in the model. The remainder is the reference map's coverage of chemistry the model does not carry, and it is drawn faint rather than dropped, since the panel is about where the measured chemistry sits in metabolism rather than about the model's boundaries. The model's own transport, exchange and lipid-chain reactions, 1,110, 274 and 188 of its 4,131, have no place on a map of chemistry and are part of what Fig. 5b carries instead.

The route. The colored modules are KEGG's gene lists for glycolysis and gluconeogenesis (`sce00010`), pyruvate metabolism (`sce00620`), the citrate cycle (`sce00020`), fatty acid degradation (`sce00071`), fatty acid biosynthesis (`sce00061`), fatty acid elongation (`sce00062`) and unsaturated fatty acid biosynthesis (`sce01040`), the last three sharing one color. A gene on more than one list takes the first list it appears on, in that order, and the order is load-bearing in both directions. Degradation has to outrank the biosynthetic lists because KEGG assigns POX1, the acyl-CoA oxidase the chassis deleted, to the unsaturated fatty acid list as well, and ordered last it takes the biosynthetic color and the whole beta-oxidation arm goes with it. Degradation must not outrank central carbon, because KEGG's degradation list also carries the alcohol and aldehyde dehydrogenases, and above glycolysis it takes twenty of that module's lines with it.

Species the map does not draw. Four of the five measured species have a compound circle on the map. Oleate has none, as the free acid or as either C18 acyl-CoA, so it is attached along the shortest path of the same genome-scale model from a compound the map does draw. That path is computed on the pathway subgraph of the model with currency metabolites removed and with substrates joined to products only, so a shared cofactor cannot make two species adjacent; every anchor at the minimal path length is kept as a candidate and the choice among them is made by the placement search below. Oleate resolves to stearate by way of stearyl-CoA and oleoyl-CoA, and the genes of that path, FAA1, FAA2, FAA4 and OLE1, are printed under its name. An attached species is drawn as an open ring so it cannot be read as a node of the map.

Where the labels go. Every drawn element is sampled onto a half-millimetre grid, the route weighted an order of magnitude above the faint background, so that a query returns how much ink lies under a rectangle or along a line. A label tries sixteen directions at four leader lengths from its node and takes the candidate whose plate and leader cover the least ink. A candidate is rejected outright if the plate leaves the panel, overlaps a placed plate, covers a labeled node, sits on a leader already drawn, or if its leader passes through a placed plate or crosses another leader. A crowded group is placed in the best of all orders rather than greedily, since a fixed order strands the last label of a cluster. The attached species go into the window of the map whose rings and links cover the least ink, scored over both sides of the column. Labels may use a 5 mm gutter to the left of the drawing, which the drawing itself may not, because the compounds this panel names sit at the left end of the fatty acid comb. Three labels have their search narrowed after review, pyruvate to upward directions and palmitate and palmitoyl-CoA to the lower right; the search still chooses among what is left, and each narrowing is recorded in the script. Label text is placed by its baseline from the typeface's cap height and descender, so the plate around a label is padded evenly around its ink rather than around its line box.

What the caption's counts are. Every count in the caption of Fig. 5a and in the panel's own key is computed by the run that draws the panel and written to `results/kegg_map/provenance.json`, together with the sha256 of the KGML, of the model file and of every stored response, the resolved systematic name of each of the twenty-three named genes, and the number of map lines each one draws. That the ten deleted regulators draw nothing is read off the KGML's own gene entries rather than asserted: none of the ten appears on any line, and all thirteen pathway genes do.

The earlier version of this panel. The same panel was first drawn on iPath3's reference map, which is the map Wu et al. use, requested from the iPath3 web service with a selection of KEGG identifiers. That drawing is older and thinner than KEGG's current one: probing it one identifier at a time showed it has no node for palmitoleate, stearate or oleate, as free acids or as acyl-CoAs, and draws nothing for ELO1, ELO2, ELO3 or OLE1, so four of the thirteen

pathway genes and three of the five measured species were missing from it. The KGML redraw carries all thirteen genes and four of the five species. The iPath3 client is kept at `experiments/008-xue-ffa/scripts/ffa_ipath_map.py` with its stored responses, and the two scripts share the label placement module `map_labels.py`.

Figures

Every panel is produced by a committed script and written as a true-size SVG at one of the standard panel widths, then converted to PDF for this document. Panels come from `perspective_figure_panels.py`, `notable_interaction_selection.py` and `ffa_epistatic_path_panels.py`; the supplementary panels and the per-readout count table behind Supplementary Note 1 and Supplementary Note 2 come from `supplementary_figure_panels.py`; the network overlay comes from `ffa_network_overlay_panel.py`, and `ffa_network_overlay_drawio.py` writes the same layout as native draw.io shapes, which is the source Fig. 5 is exported from; the growth-model figure comes from `experiments/010-kuzmin-tmi/scripts/ffa_panel_model_score.py`. The two schematic panels that open Fig. 1 and the schematic and table of Fig. 3 are native draw.io shapes written by `perspective_motivation_drawio.py` and `epistasis_model_intuition_drawio.py`, with every expression in them typeset by `epistasis_model_intuition_panels.py` and placed as an image at its measured size. The remaining multi-panel arrangements are assembled from the panel SVGs by `build_perspective_drawio.py` into editable draw.io sources under `notes/assets/drawio/`.

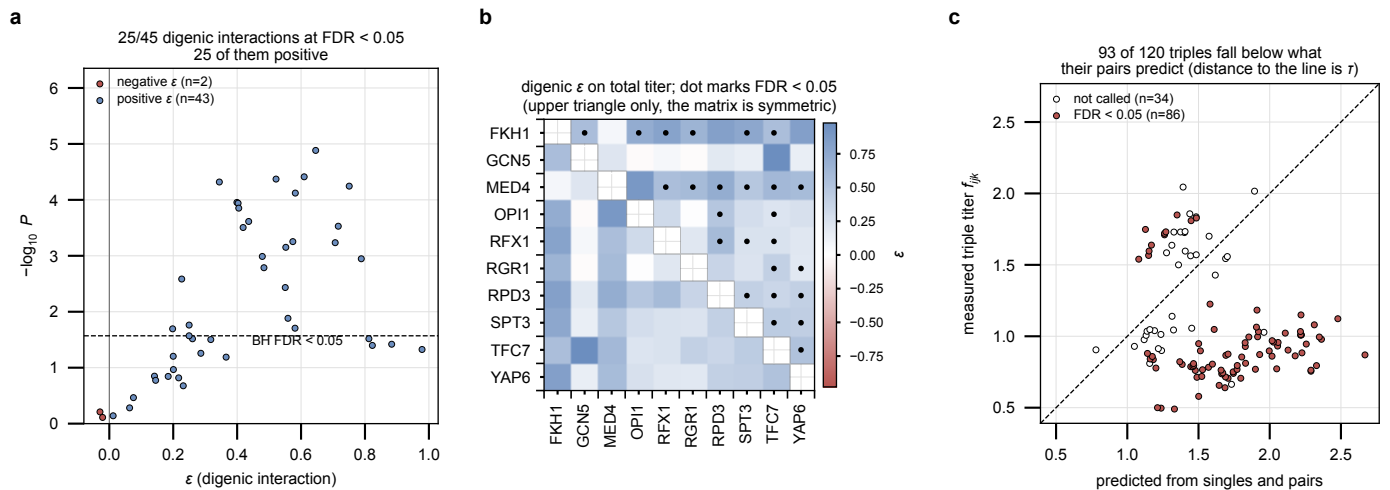
Code and data availability

All analysis scripts are in `experiments/008-xue-ffa/scripts/` and all derived tables in `experiments/008-xue-ffa/results/` of the torchcell repository. Selection of which interactions appear in a figure is itself a committed artifact rather than a post-hoc choice: `notable_interaction_selection.py` states the rule, applies it, and writes the selected set to `results/notable_interaction_selection.csv`.

Supplementary Information x

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Supplementary Figure 1 The digenic layer on total fatty acid titer. **a**, Digenic score ϵ against evidence for the 45 double deletions under the multiplicative model. The dashed line marks the largest P that Benjamini-Hochberg rejects at a 5% false discovery rate over the 45 tests of this readout; 25 pairs clear it and all 25 are positive. **b**, Every pair's ϵ as a matrix over the ten factors, blue positive and brick negative on a scale centered at zero. The matrix is symmetric and the dots marking a false-discovery-rate call are drawn in the upper triangle only. **c**, Each triple's measured titer against the titer predicted from its singles and its pairs, $f_i f_j f_k + \epsilon_{ij} f_k + \epsilon_{ik} f_j + \epsilon_{jk} f_i$. The vertical distance to the diagonal is the trigenic score τ , so a point below the line is a triple that produces less than its own pairs predict; 93 of 120 do. Filled points are triples called at a 5% false discovery rate within total titer.

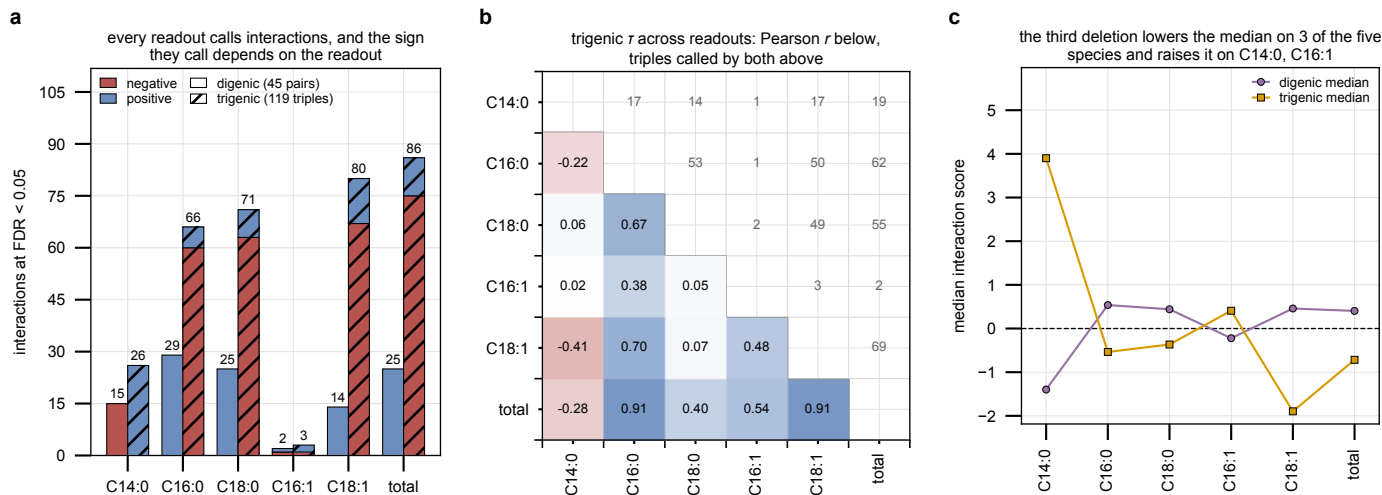
Supplementary Note 1: The digenic layer the trigenic score is measured against $\times$

The main text is a claim about the third deletion. That claim is made against what the first two do, so what the pairs do is worth stating on its own, and it is the opposite of the trigenic result.

The pairs are positive. On total titer, 43 of the 45 double deletions have a positive multiplicative digenic score and the median is +0.40. Twenty-five of the 45 reach a 5% false discovery rate within the readout, and every one of the 25 is positive (Supplementary Fig. 1a). A pair of deletions therefore tends to produce more titer than the product of its singles, which is the behavior a campaign would want and would find if it stopped at two.

No factor carries it. Drawing all 45 scores as a matrix over the ten factors (Supplementary Fig. 1b) shows the positive scores spread across the panel rather than concentrated in the rows of one or two factors. The largest score in absolute value is 0.98. Every one of the ten factors takes part in at least one called pair, from GCN5 in 1 of its 9 to TFC7 in 8 of its 9, so no single factor carries the positive digenic signal and none is exempt from it.

The third deletion undoes it. Writing the multiplicative expectation for a triple with the pairs included, as in Eq. 3, gives a prediction of the triple from its singles and its pairs alone. Ninety-three of the 120 triples fall below that prediction (Supplementary Fig. 1c), and the vertical distance to the diagonal in that panel is τ . The negative trigenic median reported in the main text is therefore not a continuation of a downward trend the pairs already set. The pairs trend upward, and the third deletion reverses the direction.



Supplementary Figure 2 The five measured species behind the summed total. **a**, Interactions called at a 5% false discovery rate within each readout, digenic (solid) and trigenic (hatched), stacked by sign, blue positive and brick negative. Counts above each bar are the totals. **b**, Trigenic τ compared across readouts over the same 119 triples: Pearson r below the diagonal, and above it the number of triples called at a 5% false discovery rate in both readouts. **c**, Median interaction score per readout for both orders. The gap between the two lines is the quantity the main text reports on total titer, and its sign is not the same on every species.

Supplementary Note 2: The five species under the summed total $\times$

Total titer is the sum of five measured fatty acids, and summing is a modeling choice in the same sense that the null is. The five species were measured separately, so the choice can be audited rather than argued about.

Every species interacts, and they do not agree on the sign. Recomputing the false-discovery-rate correction within each readout gives 3 to 80 called trigenic interactions per species against 86 on the summed total (Supplementary Fig. 2a). C16:0, C18:0 and C18:1 behave like the total, with 60 to 67 of their calls negative. C14:0 does not: all 26 of its calls are positive and its median trigenic score is +3.90, the largest of any readout. C16:1 is nearly silent, with 2 digenic and 3 trigenic calls, so a claim about it would be a claim about a readout this design cannot resolve.

C14:0 reverses between orders. On C14:0 all 15 digenic calls are negative with a median of -1.40 , and all 26 trigenic calls are positive with a median of $+3.90$ (Supplementary Fig. 2c). The direction of the interaction on that species depends on how many deletions are counted. Three of the five species move down from the digenic to the trigenic median and two move up, so the reversal the main text reports on the summed total is a majority behavior rather than a universal one.

The summed total is not a faithful summary. Correlating τ across readouts over the same 119 triples (Supplementary Fig. 2b) puts the total close to C16:0 and C18:1 at $r = 0.91$, further from C16:1 and C18:0 at $r = 0.54$ and 0.40 , and on the other side of zero from C14:0 at $r = -0.28$. The two most negatively related species are C14:0 and C18:1 at $r = -0.41$. Summing therefore reports the species that carry the most mass and cancels a species whose interactions run the other way, which is the mechanism behind the main text's observation that 31 positive consensus interactions exist across the five species and none survives on their sum.

What follows for a campaign. A campaign that optimizes one species is not optimizing a scaled version of the total, and the interaction structure it would have to navigate is not the one reported here. The reachability result of the main text is stated on total titer, and repeating it per species is a separate analysis this document does not carry out.

Supplementary Note 3: A minimal flux model of the regulator-against-enzyme comparison x

The Discussion says the comparison this design cannot make is an enzyme panel of the same shape on the same readout. That comparison can be attempted in a model, and attempting it is worth reporting, because what the model does is not what it was asked.

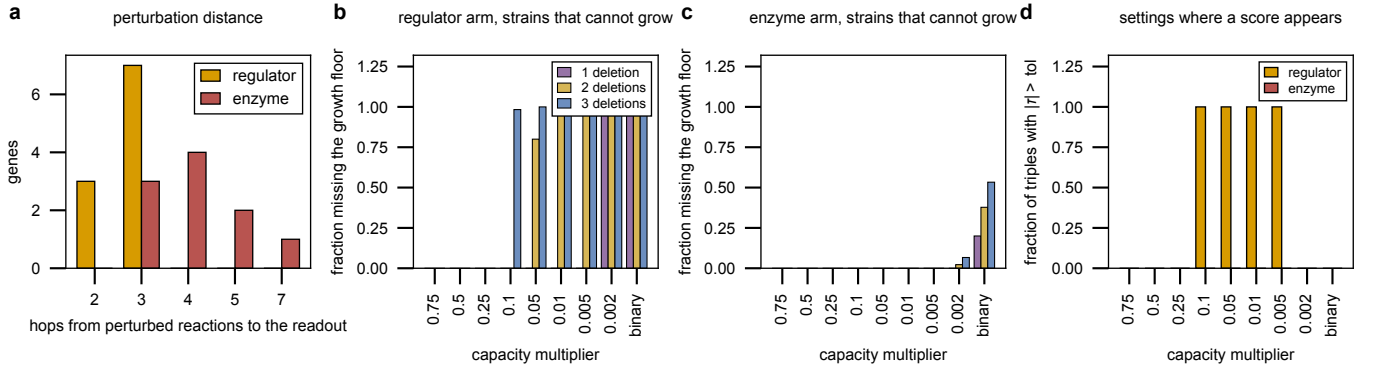
The model. The network is Yeast9, with the chassis deletions POX1, FAA1 and FAA4 applied first. The readout is the maximum of the summed export flux of the five measured species with growth held at half the base strain's maximum growth, divided by the base strain's value, so the base strain reads 1 as it does throughout this document. Two arms of ten genes are perturbed. The regulator arm is the ten deleted factors, none of which is a gene of the model, so each acts through its targets in the SGD regulatory graph and TFLink, taken in the factor to target direction and restricted to genes of the model. The enzyme arm is the ten of the thirteen pathway genes with the most reactions in the model, each acting on the reactions it catalyzes. A deletion multiplies both bounds of every reaction in its set by a capacity multiplier, or sets them to zero in the binary arm, and on a reaction hit by two deletions the multipliers multiply. The perturbation model therefore contains no interaction term of any kind, so anything the readout shows is produced by the network and by the choice to measure one locus. Interaction scores are Eqs. 1 and 2 on that readout.

At the multipliers a reader would try first, nothing happens. At 0.75, 0.5 and 0.25 every digenic and every trigenic score in both arms is zero to numerical tolerance, 2×10^{-15} against a tolerance of 10^{-6} . About seven in ten of the model's flux bounds sit at its ± 1000 placeholder rather than at a measured capacity, and scaling a placeholder binds nothing. This is a property of the model's parameterization, not a result about composability.

Below that the readout falls off a cliff rather than responding. At 0.1, 0.05, 0.01 and 0.005 the regulator arm scores a nonzero τ on all 120 triples and the enzyme arm on none, 480 against 0 over the whole sweep. Of those 480, two are graded, meaning that all seven readouts the score is computed from clear the growth floor. In the rest the score is arithmetic on zeros: τ is exactly -1 at 0.1, where the triples alone fail to grow, and exactly $+2$ at 0.05 and 0.01, where the pairs fail too. At 0.005 it is exactly twice the product of the three singles, for the same reason. The only nonzero enzyme-arm score anywhere is FAS1 with FAS2, the two subunits of fatty acid synthase, which share reactions so their multipliers compound.

The distance part of the hypothesis fails in this model. Counting hops on the model's reaction-metabolite graph with currency metabolites dropped, the regulator arm sits a median of 3 reactions from the five readout exchanges and the enzyme arm a median of 4. The reason is mechanical: a regulator's reaction set is the union over 125 to 605 model genes, 323 to 1,401 reactions, and a union that large nearly always contains something adjacent to the fatty acid exchanges, while an enzyme acts on 2 to 18. Within the regulator arm the rank correlation between a triple's mean hop count and $|\tau|$ changes sign across settings, from $\rho = -0.33$ at 0.05 to $\rho = +0.52$ at 0.01, so there is no stable relation to report.

What this establishes. Not which arm interacts more. It establishes what a model would need to answer that: graded capacities rather than placeholder bounds, a readout that does not collapse onto a feasibility boundary, and a map from a factor deletion to a distribution of expression changes rather than an all-or-nothing scaling of every target. The last of these is the step the Discussion identifies as missing, and this is what its absence costs. Everything in this note is a property of the model defined in `experiments/008-xue-ffa/scripts/regulator_vs_enzyme_epistasis_model.py`; no number in it is a titer, and none of it is evidence about the measured strains. The model also ignores whether a factor activates or represses its target, treats every target equally, and reads an optimum rather than a titer, and each of those would have to be fixed before the comparison means anything.



Supplementary Figure 3 A minimal flux model does not settle the regulator-against-enzyme comparison. **a**, Distance from the perturbed reactions to the five readout exchanges, counted in reactions on the model's reaction-metabolite graph with currency metabolites dropped. The regulator arm is closer than the enzyme arm, not further, because a regulator's reaction set is the union over hundreds of targets. **b**, The fraction of regulator-arm combinations at each order that miss the growth floor and export nothing, per capacity multiplier; every nonzero interaction score in that arm is arithmetic on these zeros. **c**, The same for the enzyme arm, which misses the floor only at 0.002 and at the binary setting, and at no order completely. **d**, The fraction of the 120 triples with $|\tau|$ above a numerical tolerance. The regulator arm scores on every triple in a four-setting window and on none outside it, and the enzyme arm scores nowhere; two of the 480 regulator-arm scores are computed entirely from strains that clear the growth floor.

Supplementary Note 4: How far a perturbation reaches on the annotation graphs ✖

The flux model of Supplementary Note 3 tested distance. The other half of the hypothesis in the Discussion is reach: a factor deletion touches many genes, each of those touches reactions, and most of those reactions are nowhere near the one locus the titer measures, so one measurement sums many small unmeasured changes. Reach can be counted on the graphs this document already uses, and the count is reported here for what it is, a property of the annotation rather than a measurement of the cell.

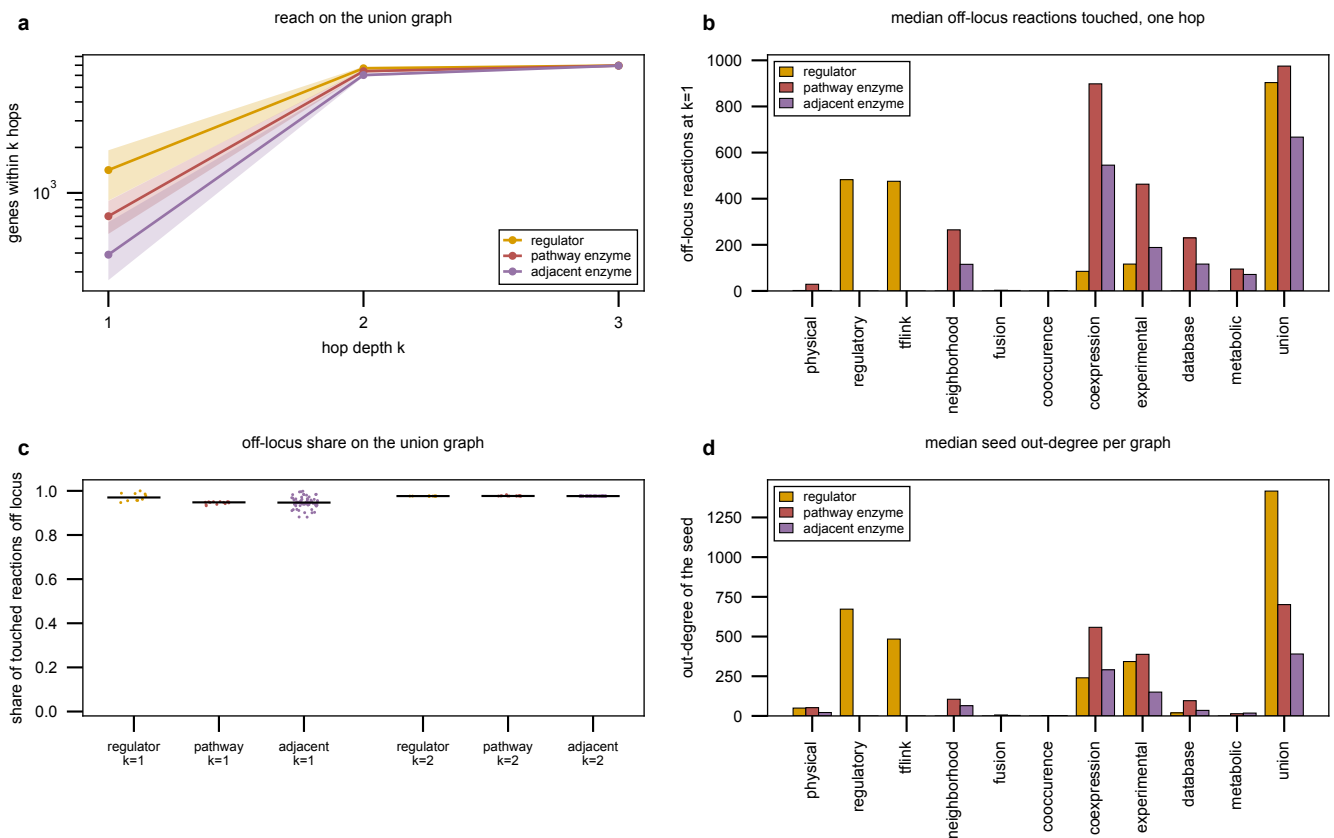
Three kinds of seed. The regulator class is the ten deleted factors, none of which is a gene of Yeast9. The pathway enzyme class is the thirteen core fatty acid genes of Fig. 5b, a direct hit on the measured locus. The adjacent enzyme class is every Yeast9 gene outside those thirteen whose reactions share at least one non-currency metabolite with a reaction a pathway gene catalyzes: one reaction away from the pathway, which is where a campaign that blocks a competing flux to divert it toward the product makes its deletions. Computed on the model with the fourteen currency metabolites of Supplementary Note 3 dropped, that class has 62 genes, sharing 113 metabolites with the pathway. The measured locus is the 69 model reactions the thirteen genes catalyze plus the five fatty acid exchanges; any other reaction is a place the titer does not look.

Ten graphs and their union. Nine are gene graphs, all but one of the ten the enrichment of Fig. 4b scans: physical association, the SGD regulatory graph, TFLink, and the six STRING 12.0 channels. The genetic interaction graph is left out, since a genetic interaction is a measured epistasis and using it here would answer the question with its own answer. The regulatory and TFLink graphs are directed and are followed factor to target only. The tenth is the model's own gene-reaction-metabolite graph with the same currency metabolites dropped, on which one gene hop is gene to reaction to metabolite to reaction to gene. The union carries an edge wherever any of the ten does. For every seed, every graph and hop depths one to three, the count is of genes reached, of model reactions those genes catalyze, and of how many of those reactions lie off the locus.

On the union the classes do not separate, and two hops reach everything. At one hop a factor reaches a median of 1,416 genes, a pathway enzyme 701 and an adjacent enzyme 390, but the reactions those genes carry outside the locus are 904, 975 and 667 respectively, and neither regulator comparison separates (Mann-Whitney $P = 0.16$ against the pathway enzymes and 0.09 against the adjacent ones). The share of touched reactions that lie off the locus is 0.97, 0.95 and 0.95. At two hops the median class reaches essentially every gene-annotated reaction the model has outside the locus, 2,645 for the regulators and the pathway enzymes and 2,644 for the adjacent enzymes, so a two-hop comparison is a comparison of ceilings rather than of reach. At three hops the class medians run from 6,950 to 6,961 of the union's 6,997 genes (Supplementary Fig. 4a, c).

What separates them is the graph, not the amount. The factors reach off-locus reactions through the two regulatory graphs only, a median of 483 through the SGD graph and 476 through TFLink at one hop, and through the metabolic graph not at all, because none of them is a gene of the model. The two enzyme classes are the mirror image: nothing through either regulatory graph, and their reach, quoted as pathway enzyme against adjacent enzyme, carried by STRING coexpression (898 and 546), experimental association (463 and 189), neighborhood (265 and 116), database (230 and 117) and the metabolic graph (95 and 72) (Supplementary Fig. 4b, d). Twenty-four of the 99 graph-by-class-by-depth cells have a median reach of zero.

What this establishes. That on these graphs a factor is not further-reaching than an enzyme in the sense the hypothesis needs; the union of curated graphs is dense enough that one hop from anything reaches a quarter of the model's reactions and two hops reach all of them. It does not establish that reach is irrelevant to the readout, because a gene one hop away on a coexpression graph need not change when the seed is deleted, no size of change is modeled, and the graphs are unevenly curated: a factor with a long TFLink list differs from one with a short list in what the resource recorded as well as in what the cell does. Every number in this note is produced by `experiments/008-xue-ffa/scripts/perturbation_reach_analysis.py` from the graphs and the model as stored, and none of it is a titer.



Supplementary Figure 4 Reach on the annotation graphs does not separate a factor from an enzyme. **a**, Genes within one, two and three hops of a seed on the union of the ten graphs, median per class with the interquartile range shaded, on a log axis. Every class saturates at two hops. **b**, Median number of Yeast9 reactions outside the measured locus carried by the genes reached at one hop, per graph and per class. The factors reach through the two regulatory graphs and the enzymes through coexpression, association and the metabolic graph; on the union the three medians are within a factor of 1.5. **c**, The share of touched reactions that lie off the locus, per seed, at one and two hops on the union. **d**, Median out-degree of a seed per graph, the first-hop fan-out behind **b**.

Supplementary Note 5: What each deleted factor costs on its own, in the growth screens x

Every titer here is a volumetric measurement with no biomass column, so a deletion that slows growth and a deletion that lowers output per cell read the same. What each of the ten factors does to growth on its own is measured elsewhere, in the genetic interaction screens, and putting those values beside the panel says which deletions bring a growth cost into the readout.

Where the numbers come from. Single-mutant fitness relative to wild type, from the strain-level tables of Costanzo et al. [2] at both assay temperatures and from the array single-mutant columns of Kuzmin et al. [4, 12]. The Costanzo uncertainty is a bootstrap standard error over 17 control screens, and the Kuzmin array columns carry no uncertainty at all, so none is quoted. Essentiality is the mirrored SGD record, counting a null allele scored inviable. The lookup, every strain it found and the file and column each value was read from are in `experiments/008-xue-ffa/scripts/tf_single_mutant_fitness.py` and its CSV; the table below is one row per gene, and it names an allele class only where the classes disagree.

Gene	ORF	Screen allele	Costanzo 26 °C	Costanzo 30 °C	Kuzmin
FKH1	YIL131C	KanMX	1.01	1.01	1.01
GCN5	YGR252W	no strain	—	—	—
MED4	YOR174W	ts	0.52	0.42	0.53 inviable
OPI1	YHL020C	KanMX, NatMX	0.85	0.85	0.85
RFX1	YLR176C	KanMX, NatMX	0.97	0.97	0.95
RGR1	YLR071C	DAmP, ts	0.97 (DAmP), 0.50 (ts)	0.97 (DAmP), 0.53 (ts)	0.50 inviable
RPD3	YNL330C	NatMX	—	—	—
SPT3	YDR392W	KanMX, NatMX	0.73	0.73	0.72
TFC7	YOR110W	DAmP	—	—	— inviable
YAP6	YDR259C	KanMX, NatMX	1.00	1.00	1.00

Table 1 Single-mutant fitness of the ten deleted transcription factors in the growth screens, relative to wild type. A dash is a gene the screen carries no fitness value for, either because it has no strain there or because the released table leaves the value empty. “inviability” marks a gene whose null allele SGD records as inviable, which is why the screens carry a temperature-sensitive or DAmP allele for it rather than a deletion.

Three of the ten are not deletable in a laboratory background. MED4, RGR1 and TFC7 all carry an inviable null in SGD, and no screen has a deletion strain for any of them: Costanzo represents MED4 by temperature-sensitive alleles, RGR1 by a DAmP allele and a temperature-sensitive one, and TFC7 by a DAmP allele with no fitness reported. Where the hypomorphs do carry a value they are the sickest strains in the set, 0.52 for MED4 and 0.50 for the RGR1 temperature-sensitive allele against 0.97 for the RGR1 DAmP allele, which is the difference between two perturbation strengths rather than two estimates of one quantity. The source study nonetheless reports viable deletion strains for all three in its producing background. Reconciling that is outside what these data can do, and the possibilities, a different background, a partial deletion, or a suppressor, are not distinguishable from the released tables. What follows for this document is narrower: for these three factors the titer of a strain carrying them is not comparable to a laboratory-background deletion of the same gene.

The rest are mild, and one is absent. FKH1 and YAP6 are at or within a percent of wild-type fitness, RFX1 at 0.97, OPI1 at 0.85 and SPT3 at 0.73, the largest measured cost among the seven non-essential factors. RPD3 has a strain in Costanzo but no reported fitness, and GCN5 has no strain in any of the five tables, checked both by systematic name and by allele name. GCN5 is the one factor whose single deletion raises titer here, so the screens offer no growth value for the single most useful deletion in the panel.

What this does and does not license. It does not convert any titer into a per-cell quantity, and it is not a measurement in the producing background. It bounds one reading of the results: the two single deletions that cost the most titer, RPD3 at 0.52× and MED4, are a gene with no reported growth value and a gene whose null is inviable in a laboratory strain, so a growth contribution to those two titers cannot be ruled out from these data. The factors with mild growth costs, FKH1, YAP6 and RFX1, lose titer in this chassis anyway, which is a titer effect rather than a growth one.

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